Chapter 12: Spin and Statistics

Quantum Field Theory and the Standard Model

Matthew D. Schwartz

Injung Kim

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Introduction: The Spin-Statistics Theorem

Core Physical Statement

- **Integer spin particles** (spins $s = 0, 1, 2, \ldots$) are symmetric under particle interchange $\rightarrow$ **Bosons** (Bose–Einstein statistics).
- **Half-integer spin particles** (spins $s = \frac{1}{2}, \frac{3}{2}, \ldots$) are antisymmetric under particle interchange $\rightarrow$ **Fermions** (Fermi–Dirac statistics).

The Four Complementary Derivations in Chapter 12

1. **Path Dependence & Topology** (Section 12.2): Rotating identical particles by π.
2. **Lorentz Invariance of the S-matrix** (Section 12.4): Time-ordered products in propagators.
3. **Stability** (Section 12.5): Boundedness from below of the total energy $E = \int d^3x\, T^{00}$.
4. **Causality** (Section 12.6): Vanishing of field commutators/anticommutators at spacelike separation $(x-y)^2 < 0$.

12.1 Identical Particles — Multi-Particle States

Definition of Identical Particles

- Particles transforming in the same representation of the Poincaré group and sharing identical internal quantum numbers n_i (momentum $\vec{p}_i$, spin s_i, species n).

Multi-Particle State Definition (Eq. 12.2 & 12.4)

$$|\dots s_1\vec{p}_1 n \dots s_2\vec{p}_2 n \dots\rangle = \dots \sqrt{2\omega_1}\, a^\dagger_{\vec{p}_1 s_1 n} \dots \sqrt{2\omega_2}\, a^\dagger_{\vec{p}_2 s_2 n} \dots |0\rangle$$

$$|\dots s_2\vec{p}_2 n \dots s_1\vec{p}_1 n \dots\rangle = \dots \sqrt{2\omega_2}\, a^\dagger_{\vec{p}_2 s_2 n} \dots \sqrt{2\omega_1}\, a^\dagger_{\vec{p}_1 s_1 n} \dots |0\rangle$$

Variables & Notation

- $a^\dagger_{\vec{p}_i s_i n}$: Creation operator for a particle of species n, spin s_i, and 3-momentum $\vec{p}_i$.
- $\omega_i = \sqrt{\vec{p}_i^{\,2} + m^2}$: Relativistic on-shell energy of particle i.
- $|0\rangle$: The invariant vacuum ground state.

12.1 Identical Particles — Exchange Phase

Exchange Phase Condition (Eq. 12.5)

- Since particles are indistinguishable, swapping two identical particles must yield the exact same physical state up to an unobservable global phase factor $\alpha_n = e^{i\phi}$:

$$|\ldots s_1\vec{p}_1 n \ldots s_2\vec{p}_2 n \ldots\rangle = \alpha_n |\ldots s_2\vec{p}_2 n \ldots s_1\vec{p}_1 n \ldots\rangle$$

Double Interchange Constraint (Eq. 12.6)

- Performing the interchange twice returns the system to its initial configuration:

$$|\ldots s_1\vec{p}_1 n \ldots s_2\vec{p}_2 n \ldots\rangle = \alpha_n^2 |\ldots s_1\vec{p}_1 n \ldots s_2\vec{p}_2 n \ldots\rangle$$

$$\implies \alpha_n^2 = 1 \implies \alpha_n = \pm 1$$

- Only two quantum statistics options exist in 3D: $\alpha_n = +1$ (Bosons) and $\alpha_n = -1$ (Fermions).

12.1 Identical Particles — Bosons & Commutators

Bosonic Statistics ($\alpha_n = +1$)

- States are **symmetric** under the interchange of any two identical particles.
- The creation and annihilation operators satisfy canonical **commutation relations**:

$$[a_{\vec{p}_1 s_1 n}, a^\dagger_{\vec{p}_2 s_2 n}] = (2\pi)^3 \delta^3(\vec{p}_1 - \vec{p}_2)\delta_{s_1,s_2} \quad \text{(Eq. 12.8)}$$

$$[a^\dagger_{\vec{p}_1 s_1 n}, a^\dagger_{\vec{p}_2 s_2 n}] = 0 \quad \text{(Eq. 12.9)}$$

$$[a_{\vec{p}_1 s_1 n}, a_{\vec{p}_2 s_2 n}] = 0$$

Physical Properties of Bosons

- Arbitrary numbers of bosons can occupy the same quantum state.
- Associated with integer spins ($s = 0, 1, 2, \ldots$) in quantum field theory.

12.1 Identical Particles — Fermions & Anticommutators

Fermionic Statistics ($\alpha_n = -1$)

- States are **antisymmetric** under the interchange of any two identical particles.
- The creation and annihilation operators satisfy canonical **anticommutation relations**:

$$\{a_{\vec{p}_1 s_1 n}, a^\dagger_{\vec{p}_2 s_2 n}\} = (2\pi)^3 \delta^3(\vec{p}_1 - \vec{p}_2)\delta_{s_1,s_2} \quad \text{(Eq. 12.10)}$$

$$\{a^\dagger_{\vec{p}_1 s_1 n}, a^\dagger_{\vec{p}_2 s_2 n}\} = 0 \quad \text{(Eq. 12.11)}$$

$$\{a_{\vec{p}_1 s_1 n}, a_{\vec{p}_2 s_2 n}\} = 0$$

Definition of Anticommutator

$$\{A, B\} \equiv AB + BA$$

12.1 Identical Particles — Pauli Exclusion Principle

Derivation of Pauli Exclusion Principle (Eq. 12.12)

- Applying the anticommutator $\{a^\dagger_{\vec{p}sn}, a^\dagger_{\vec{p}sn}\} = 0$ to the vacuum state $|0\rangle$:

$$a^\dagger_{\vec{p}sn} a^\dagger_{\vec{p}sn}|0\rangle = -a^\dagger_{\vec{p}sn} a^\dagger_{\vec{p}sn}|0\rangle = 0$$

- **Physical Conclusion**: No two identical fermions can occupy the exact same single-particle quantum state.

Cluster Decomposition & Species Independence

- By the cluster decomposition principle, all particles of the same species n must share the exact same phase $\alpha_n = \pm 1$.

12.2 Path Dependence — Topological Exchange Operator

Topological Exchange Definition (Eq. 12.13)

- Define a continuous topological exchange operator S moving particle 1 at x_1 to x_2 and particle 2 at x_2 to x_1 along a path parameterized by angle ϕ:

$$S|\phi_1(x_1)\phi_2(x_2)\rangle = e^{i\phi\kappa}|\phi_2(x_1)\phi_1(x_2)\rangle$$

Topological Unwrapping in 3D Space

- In $3+1$ dimensions, a closed path corresponding to a 2π loop around another particle can be continuously unwrapped to a point.
- Single-valuedness under a full 2π rotation requires $\kappa \in \mathbb{Z}$.
- A single exchange corresponds to a rotation by $\phi = \pi$ (Eq. 12.14):

$$S|\phi_1(x_1)\phi_2(x_2)\rangle = e^{i\pi\kappa}|\phi_2(x_1)\phi_1(x_2)\rangle = \pm|\phi_2(x_1)\phi_1(x_2)\rangle$$

12.2 Path Dependence — Kinematics of Exchange

Decomposing Particle Exchange

- Exchanging two identical particles at x_1 and x_2 is kinematically equivalent to two operations:
  i. Translating particle 1 to x_2 and particle 2 to x_1.
  ii. Rotating the entire two-particle system by π around their center of mass.

Role of Poincaré Group

- Spatial translations do not alter internal spin phase factors.
- Therefore, the exchange phase α_n is **entirely determined by how the fields transform under spatial rotations by π**!

12.2 Path Dependence — Rotation by π for Scalars

Scalar Field Transformation under Rotation

- For scalar fields ($s = 0$), spatial rotation acts trivially on the internal field components:

$$R(\pi)\phi(x)R^{-1}(\pi) = \phi(-x)$$

Exchange Phase for Scalar Fields (Eq. 12.15)

- Rotating two identical scalar particles by π around their midpoint introduces no internal spin phase factor:

$$S|\phi_1(x_1)\phi_2(x_2)\rangle = +|\phi_2(x_1)\phi_1(x_2)\rangle \quad (\text{Eq. } 12.15)$$

- **Conclusion**: Scalar fields ($s = 0$) are strictly **Bosons** ($\alpha_n = +1$).

12.2 Path Dependence — Rotation by π for Spinors

Spinor Rotation Representation (Eq. 12.16)

- For Dirac spinors ($s = 1/2$), the rotation operator around the z-axis by angle θ_z is represented by:

$$\Lambda_s(\theta_z) = \exp(i\theta_z S^{12}) = \mathrm{diag}\left(e^{i\theta_z/2}, e^{-i\theta_z/2}, e^{i\theta_z/2}, e^{-i\theta_z/2}\right)$$

Eigenvalues for Rotation by π (Eq. 12.17)

- Setting $\theta_z = \pi$:

$$\Lambda_s(\pi) = \mathrm{diag}\left(e^{i\pi/2}, e^{-i\pi/2}, e^{i\pi/2}, e^{-i\pi/2}\right) = \mathrm{diag}(i, -i, i, -i)$$

- Individual spinor components acquire eigenvalues of $+i$ or $-i$.

12.2 Path Dependence — Spinor Exchange Phase

Two-Spinor Combined Rotation (Eq. 12.18)

- When rotating a two-spinor state by π, the rotation acts on both particle states simultaneously:

$$\Lambda_s(\pi) \otimes \Lambda_s(\pi) \longrightarrow i \times i = -1$$

- The product of the two rotational phase factors yields -1:

$$S|\psi_1(x_1)\psi_2(x_2)\rangle = -|\psi_2(x_1)\phi_1(x_2)\rangle \quad (\text{Eq. } 12.18)$$

Conclusion from Path Topology

- Spin-1/2 Dirac particles are strictly **Fermions** ($\alpha_n = -1$).

12.3 Quantizing Spinors — Complex Scalar Expansion

Complex Scalar Field Operator (Eq. 12.20 & 12.21)

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left(a_p e^{-ipx} + b_p^\dagger e^{ipx}\right)$$

$$\phi^\dagger(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left(a_p^\dagger e^{ipx} + b_p e^{-ipx}\right)$$

Physical Interpretation of Operators

- $a_p^\dagger$: Creates a particle of mass m and positive electric charge $+q$.
- $b_p^\dagger$: Creates an antiparticle of mass m and negative electric charge $-q$.
- $p^\mu = (\omega_p, \vec{p})$ with $\omega_p = \sqrt{\vec{p}^2 + m^2}$.

12.3 Quantizing Spinors — Free Dirac Field Expansion

Dirac Field Operator Expansion (Eq. 12.26 & 12.27)

$$\psi(x) = \sum_{s=1}^{2} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left(a_{p,s} u_s(p) e^{-ipx} + b_{p,s}^\dagger v_s(p) e^{ipx} \right)$$

$$\bar{\psi}(x) = \sum_{s=1}^{2} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left(a_{p,s}^\dagger \bar{u}_s(p) e^{ipx} + b_{p,s} \bar{v}_s(p) e^{-ipx} \right)$$

Variables & Notation

- $\bar{\psi}(x) \equiv \psi^\dagger(x)\gamma^0$: Dirac adjoint field.
- $u_s(p)$: Positive-energy basis spinor ($(\not{p} - m)u_s(p) = 0$).
- $v_s(p)$: Negative-energy basis spinor ($(\not{p} + m)v_s(p) = 0$).
- Note: $\not{p} \equiv \gamma^\mu p_\mu$.

12.3 Quantizing Spinors — Basis Spinor Solutions

Explicit Dirac Representation Spinors (Eq. 12.25)

$$u_s(p) = \begin{pmatrix} \sqrt{p\cdot\sigma}\,\xi_s \\ \sqrt{p\cdot\bar{\sigma}}\,\xi_s \end{pmatrix}, \quad v_s(p) = \begin{pmatrix} \sqrt{p\cdot\sigma}\,\eta_s \\ -\sqrt{p\cdot\bar{\sigma}}\,\eta_s \end{pmatrix}$$

2-Component Spin Basis

- $\sigma^\mu = (\mathbf{1}, \vec{\sigma})$, $\bar{\sigma}^\mu = (\mathbf{1}, -\vec{\sigma})$.
- Constant orthonormal 2-component spinors:

$$\xi_1 = \eta_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \xi_2 = \eta_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- Normalization: $\bar{u}_r(p)u_s(p) = 2m\delta_{rs}$, $\bar{v}_r(p)v_s(p) = -2m\delta_{rs}$.

12.4 Lorentz Invariance of S-matrix — 12.4.1 Spin 0

Two-Point Vacuum Matrix Elements (Eq. 12.30 & 12.31)

- Evaluating the matrix elements of complex scalar fields between vacuum states:

$$\langle 0|\phi^\dagger(x)\phi(0)|0\rangle = \int \frac{d^3p}{(2\pi)^3}\frac{1}{2\omega_p}e^{-i\omega_p t + i\vec{p}\cdot\vec{x}}$$

$$\langle 0|\phi(0)\phi^\dagger(x)|0\rangle = \int \frac{d^3p}{(2\pi)^3}\frac{1}{2\omega_p}e^{i\omega_p t - i\vec{p}\cdot\vec{x}}$$

Role of Time-Ordering

- S-matrix elements in perturbation theory depend directly on time-ordered products of field operators.

12.4.1 Spin 0 — Propagator with Commutators

Time-Ordering Definition with Commutators (Eq. 12.32)

$$T\{\phi^\dagger(x)\phi(0)\} = \phi^\dagger(x)\phi(0)\theta(t) + \phi(0)\phi^\dagger(x)\theta(-t)$$

Contour Integration Result (Eq. 12.35)

- Combining the two theta-function integrals via complex p_0 contour integration:

$$\langle 0|T\{\phi^\dagger(x)\phi(0)\}|0\rangle = \int \frac{d^4p}{(2\pi)^4}\frac{i}{p^2 - m^2 + i\varepsilon}e^{ipx}$$

- **Physical Consequence**: The propagator is **manifestly Lorentz invariant** because $p^2 = p_0^2 - \vec{p}^2$ and d^4p are Lorentz scalars!

12.4.1 Spin 0 — Failure of Anticommutators

Hypothetical Anticommutative Time-Ordering (Eq. 12.36)

- Suppose we mistakenly quantized scalar fields using anticommutators:

$$T_{\text{anti}}\{\phi^\dagger(x)\phi(0)\} = \phi^\dagger(x)\phi(0)\theta(t) - \phi(0)\phi^\dagger(x)\theta(-t)$$

Resulting Matrix Element (Eq. 12.36)

$$\langle 0|T_{\text{anti}}\{\phi^\dagger(x)\phi(0)\}|0\rangle = \int \frac{d^4p}{(2\pi)^4} \frac{-i\omega_p}{(p^2 - m^2 + i\varepsilon)\sqrt{\vec{p}^2 + m^2}} e^{ipx}$$

- **Physical Conclusion**: The numerator contains $\omega_p = \sqrt{\vec{p}^2 + m^2}$, which is **NOT** a Lorentz scalar! S-matrix Lorentz invariance is destroyed. Spin 0 MUST use commutators.

12.4.2 Spinors — Spinor Outer Product Completeness

Outer Product Sums (Eq. 12.38 & 12.39)

- Summing over the two physical spin polarizations $s = 1, 2$:

$$\sum_{s=1}^{2} u_s(p)\bar{u}_s(p) = \not{p} + m = \gamma^\mu p_\mu + m$$

$$\sum_{s=1}^{2} v_s(p)\bar{v}_s(p) = \not{p} - m = \gamma^\mu p_\mu - m$$

Variables

- γ^μ: 4×4 Dirac matrices satisfying $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$.
- $\not{p} = \gamma^0 E_{\vec{p}} - \vec{\gamma} \cdot \vec{p}$.

12.4.2 Spinors — Two-Point Matrix Elements

Evaluating Dirac Matrix Elements (Eq. 12.41 & 12.42)

$$\langle 0|\psi(0)\bar{\psi}(x)|0\rangle = \sum_s \int \frac{d^3p}{(2\pi)^3}\frac{1}{2\omega_p} v_s(p)\bar{v}_s(p)e^{ipx} = (-i\not{\partial} + m)\int \frac{d^3p}{(2\pi)^3}\frac{1}{2\omega_p}e^{ipx}$$

$$\langle 0|\bar{\psi}(x)\psi(0)|0\rangle = \sum_s \int \frac{d^3p}{(2\pi)^3}\frac{1}{2\omega_p} u_s(p)\bar{u}_s(p)e^{-ipx} = -(-i\not{\partial} + m)\int \frac{d^3p}{(2\pi)^3}\frac{1}{2\omega_p}e^{-ipx}$$

- Note the relative minus sign between the two matrix elements!

12.4.2 Spinors — Time-Ordering with Anticommutators

Fermionic Time-Ordering Definition (Eq. 12.43)

- Due to the relative minus sign, we MUST define fermionic time-ordering with a MINUS sign for field interchange:

$$T\{\psi(0)\bar{\psi}(x)\} = \psi(0)\bar{\psi}(x)\theta(-t) - \bar{\psi}(x)\psi(0)\theta(t)$$

Dirac Propagator Result (Eq. 12.45)

$$\langle 0|T\{\psi(0)\bar{\psi}(x)\}|0\rangle = \int \frac{d^4p}{(2\pi)^4} \frac{i(\not{p}+m)}{p^2-m^2+i\varepsilon} e^{ipx}$$

- **Physical Conclusion**: The minus sign in fermionic time-ordering produces a **manifestly Lorentz invariant** Dirac propagator!

12.4.2 Spinors — Physical Origin of the Minus Sign

Spatial Rotation Analysis

- Notice the Dirac operator: $\not{p} = \gamma^0 E_{\vec{p}} - \vec{\gamma} \cdot \vec{p}$.
- The spatial component $\vec{\gamma} \cdot \vec{p}$ is **odd** under the spatial rotation by π ($\vec{p} \to -\vec{p}$).
- Half-integer spin fields pick up an extra internal phase $i \times i = -1$ under spatial rotation by π.
- This internal kinematic minus sign requires anticommutative time-ordering to preserve S-matrix Lorentz invariance.

12.5 Stability — Energy Density & Noether's Theorem

Energy Density Definition (Eq. 12.50)

- From Noether's theorem for time-translation invariance, the energy density is the 00-component of the energy-momentum tensor $T^{\mu\nu}$:

$$\mathcal{E} = T^{00} = \sum_n \frac{\partial \mathcal{L}}{\partial \dot{\phi}_n} \dot{\phi}_n - \mathcal{L}$$

Total Energy Operator (Eq. 12.51)

$$E = \int d^3x \, \mathcal{E}$$

- **Vacuum Stability Postulate**: For a physical ground state to exist, the total energy E must be **bounded from below**.

12.5.1 Complex Scalar Fields — Positive Energy

Scalar Energy Density (Eq. 12.52 & 12.53)

$$\mathcal{L} = |\partial_\mu \phi|^2 - m^2|\phi|^2 \implies \mathcal{E} = |\partial_t \phi|^2 + |\vec{\nabla}\phi|^2 + m^2|\phi|^2$$

- Classically, $\mathcal{E} \geq 0$ is manifestly positive-definite.

Quantum Energy Operator with Commutators (Eq. 12.57)

- Substituting field expansions and using $[a_p, a_q^\dagger] = (2\pi)^3\delta^3(\vec{p}-\vec{q})$:

$$E = \int \frac{d^3p}{(2\pi)^3}\omega_p \left(a_p^\dagger a_p + b_p^\dagger b_p\right) + VE_0$$

- $VE_0 = V\int \frac{d^3p}{(2\pi)^3}\omega_p > 0$: Infinite positive zero-point energy.
- All excitation states have strictly positive energy above ground state $\implies$ Stable!

12.5.1 Complex Scalars — Failure of Anticommutators

Quantization using Anticommutators (Eq. 12.59 & 12.60)

- Suppose we quantized complex scalar fields using anticommutators $\{b_p, b_q^\dagger\} = (2\pi)^3 \delta^3(\vec{p} - \vec{q})$:

$$E = \int \frac{d^3p}{(2\pi)^3} \omega_p \left(a_p^\dagger a_p - b_p^\dagger b_p \right) + V E_0$$

Instability Proof

- The antiparticle number operator $-b_p^\dagger b_p$ enters with a **negative sign**.
- Creating arbitrary numbers of antiparticles lowers the system's energy to $-\infty$.
- The vacuum instantly collapses into an infinite flux of antiparticles $\implies$ Unstable!

12.5.2 Free Fermions — Energy Operator Structure

Dirac Energy Density (Eq. 12.61 & 12.62)

$$\mathcal{L} = \bar{\psi}(i \not{\partial} - m)\psi \implies \mathcal{E} = T^{00} = i\bar{\psi}\gamma^0\partial_t\psi$$

Unintegrated Quantum Operator (Eq. 12.66)

- Substituting Dirac field expansions into the energy integral:

$$E = \sum_s \int \frac{d^3p}{(2\pi)^3}\omega_p \left(a^\dagger_{p,s}a_{p,s} - b_{p,s}b^\dagger_{p,s}\right)$$

- Notice the explicit minus sign in front of the antiparticle term $b_{p,s}b^\dagger_{p,s}$!

12.5.2 Free Fermions — Quantization with Anticommutators

Reordering with Anticommutators (Eq. 12.67)

- Using fermionic anticommutators $\{b_{p,s}, b^\dagger_{p,s}\} = (2\pi)^3\delta^3(0)$:

$$b_{p,s}b^\dagger_{p,s} = -b^\dagger_{p,s}b_{p,s} + (2\pi)^3\delta^3(0)$$

$$E = \sum_s \int \frac{d^3p}{(2\pi)^3}\omega_p \left(a^\dagger_{p,s}a_{p,s} + b^\dagger_{p,s}b_{p,s}\right) - VE_0$$

Physical Conclusions

- $-VE_0 = -2V\int \frac{d^3p}{(2\pi)^3}\omega_p$: Negative infinite zero-point energy (Dirac sea).
- Both particle ($a^\dagger a$) and antiparticle ($b^\dagger b$) excitations have **strictly positive energy** $\Longrightarrow$ Stable!

12.5.2 Free Fermions — Failure of Commutators

Reordering with Commutators (Eq. 12.68)

- If we quantization Dirac fields using commutators $[b_{p,s}, b^\dagger_{p,s}] = (2\pi)^3\delta^3(0)$:

$$b_{p,s}b^\dagger_{p,s} = b^\dagger_{p,s}b_{p,s} + (2\pi)^3\delta^3(0)$$

$$E = \sum_s \int \frac{d^3p}{(2\pi)^3}\omega_p \left(a^\dagger_{p,s}a_{p,s} - b^\dagger_{p,s}b_{p,s}\right) - VE_0$$

Instability Proof

- Energy is **unbounded from below** because creating antiparticles reduces energy indefinitely.
- Fermions MUST be quantized with anticommutators to ensure vacuum stability.

12.5.3 General Spins — Derivative Structure Origin

Kinematic Derivative Order

- **Scalar Fields ($s = 0$)**: Kinetic Lagrangian contains **two time derivatives** $(\partial_t \phi)^2$.
  - Energy density $\mathcal{E} \sim \dot{\phi}^2$ is quadratic and positive-definite.
  - Commutators preserve the positive sign for antiparticles.
- **Fermion Fields ($s = 1/2$)**: Kinetic Lagrangian contains **one time derivative** $\bar{\psi} i \gamma^0 \partial_t \psi$.
  - Energy density is linear in time derivatives.
  - Anticommutators are required to flip the minus sign of antiparticles to $+$.

12.6 Causality — Microcausality Postulate

Principle of Microcausality (Eq. 12.71)

- Physical observables $O_1(x)$ and $O_2(y)$ constructed from field operators must commute at spacelike separation:

$$[O_1(x), O_2(y)] = 0 \quad \text{for } (x-y)^2 < 0$$

Physical Meaning

- Ensures that a measurement at spacetime point x cannot influence a measurement at y if no light signal can travel between them ($r > |t|$).
- Prevents faster-than-light information transfer and causal paradoxes.

12.6 Causality — Scalar Field Commutator Distribution

Scalar Field Commutator Evaluation (Eq. 12.74 & 12.75)

$$[\phi(x), \phi^\dagger(y)] = \int \frac{d^3q}{(2\pi)^3} \frac{1}{2\omega_q} \left(e^{-iq(x-y)} - e^{iq(x-y)} \right) \equiv iD(t, \vec{r})$$

Invariant $D(t, \vec{r})$ Properties (Eq. 12.78 & 12.79)

- $(\Box + m^2) D(t, \vec{r}) = 0$
- $D(0, \vec{r}) = 0$ (Equal-time commutator vanishes)
- $\partial_t D(t, \vec{r})\big|_{t=0} = -\delta^3(\vec{r})$
- $D(t, \vec{r})$ is **odd** under time reversal ($D(-t, \vec{r}) = -D(t, \vec{r})$).
- **Crucial Result**: $D(t, \vec{r}) = 0$ for all spacelike intervals $(x-y)^2 < 0$!

12.6 Causality — Failure of Anticommutators for Scalars

Scalar Anticommutator Distribution (Eq. 12.81 & 12.82)

$$\{\phi(x), \phi^\dagger(y)\} = \int \frac{d^3q}{(2\pi)^3} \frac{1}{2\omega_q} \left(e^{-iq(x-y)} + e^{iq(x-y)}\right) \equiv iD_1(t, r)$$

Massless $D_1(t, r)$ Expression (Eq. 12.83)

- For massless scalar fields ($m = 0$):

$$D_1(t, r) = \frac{1}{2\pi^2} \frac{1}{r^2 - t^2}$$

- $D_1(t, r)$ is **even** under time reversal ($D_1(-t, r) = D_1(t, r)$).
- **Physical Conclusion**: $D_1(t, r) \neq 0$ outside the lightcone ($r > t$). Quantizing scalars with anticommutators **violates microcausality**!

12.6.1 Causality — Spinor Fields

Dirac Field Commutator (Eq. 12.87)

- If Dirac fields are quantized with **commutators**:

$$[\psi(x), \bar{\psi}(y)] = (i \not{\partial}_x + m) D_1(t, r) \neq 0 \quad \text{for } (x-y)^2 < 0$$

- Non-zero spacelike commutator $\implies$ Violates Microcausality!

Dirac Field Anticommutator (Eq. 12.89)

- If Dirac fields are quantized with **anticommutators**:

$$\{\psi(x), \bar{\psi}(y)\} = (i \not{\partial}_x + m) D(t, r) = 0 \quad \text{for } (x-y)^2 < 0$$

- **Physical Conclusion**: Bilinear observables $[\bar{\psi}\psi(x), \bar{\psi}\psi(y)] = 0$ commute spacelike $\implies$ Preserves Microcausality!

12.6.2 Higher Spins & Pauli's PT Symmetry Argument

Massive Vector Fields ($s = 1$) Commutator (Eq. 12.92)

$$[A_\mu(x), A_\nu(y)] = \left(g_{\mu\nu} + \frac{1}{m^2}\partial_\mu\partial_\nu\right) D(t, r) \implies \text{Requires Commutators}$$

Transformation under PT (Eq. 12.93 & 12.94)

- Combined Parity (P) and Time Reversal (T):

$$D(t, \vec{r}) = -D(-t, -\vec{r}) \quad (\text{odd under } PT)$$

$$D_1(t, \vec{r}) = D_1(-t, -\vec{r}) \quad (\text{even under } PT)$$

General Derivative Rule

- **Integer Spins**: Even number of derivatives $\rightarrow$ linked to $D(t, r) \rightarrow$ **Commutators**.
- **Half-Integer Spins**: Odd number of derivatives $\rightarrow$ linked to $\not{\partial} D(t, r) \rightarrow$ **Anticommutators**.

Comprehensive Summary Table

Property	**Integer Spin ($s = 0, 1, 2$)**	**Half-Integer Spin ($s = 1/2, 3/2$)**
Statistics	Bose–Einstein ($\alpha_n = +1$)	Fermi–Dirac ($\alpha_n = -1$)
Quantization	Commutators $[a_p, a_q^\dagger]$	Anticommutators $\{a_p, a_q^\dagger\}$
Time-Ordering	$T\{\phi_1\phi_2\} = \phi_1\phi_2\theta + \phi_2\phi_1\theta$	$T\{\psi_1\psi_2\} = \psi_1\psi_2\theta - \psi_2\psi_1\theta$
Propagator	$\frac{i}{p^2 - m^2 + i\varepsilon}$ (Lorentz Invariant)	$\frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon}$ (Lorentz Invariant)
Zero-Point Energy	$+VE_0$ (Positive)	$-VE_0$ (Dirac Sea)
Energy Spectrum	Bounded from below	Bounded from below
Causality	$D(t, r) = 0$ spacelike	$(i\ \not{\partial} + m)D(t, r) = 0$ spacelike

Fundamental Conclusions

Summary of Chapter 12 Foundations

1. **Path Topology**: Rotation by π generates an internal phase factor of $(-1)^{2s}$.
2. **Lorentz Invariance**: Time-ordered S-matrix elements are Lorentz invariant if and only if spin-statistics holds.
3. **Stability**: Total field energy is bounded from below if and only if integer spins use commutators and half-integer spins use anticommutators.
4. **Microcausality**: Physical observables commute at spacelike separation $(x-y)^2 < 0$.

All four independent theoretical frameworks yield the exact same physical result: the **Spin-Statistics Theorem**.